INCH-POUND

MIL-R-23417/9A
25 April 1991

SUPERSEDING
MIL-R-23417/9
15 September 1967

MILITARY SPECIFICATION SHEET

RESOLVER, ELECTRICAL, AC
TYPE 15R28N4A

This specification is approved for use by all Departments and Agencies of the Department of Defense. The requirements for acquiring the Electrical Resolvers described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) cited in the solicitation: MIL-R-23417.

TABLE I. Requirements.

| Requirements | Value | Unit | Tolerance |
|---|---|---|---|
| Excitation winding | Stator | --- | --- |
| Rated frequency | 400 | Hz | ±1% |
| Input voltage: | | | |
| Maximum | 26 | Volts rms | ±1% |
| Test | 10 | Volts rms | ±1% |
| Minimum | 0.5 | Volts rms | ±1% |
| Primary current | | | |
| @ test voltage | 4.0 | milliamps | nominal |
| Primary power | 0.005 | watts | nominal |
| D.C. resistance: | | | |
| Rotor | 170 | ohms | ±15% |
| Stator | 200 | ohms | ±15% |
| Impedance: | | | |
| Zro | 440 +j2970 | ohms | R±15%+jx±10% |
| Zso | 400 +j2800 | ohms | R±15%+jx±10% |
| Zrs | 390 +j250 | ohms | Reference |
| Transformation ratio: | | | |
| Output/input | 0.980 | --- | ±1% |
| Equality of transformation ratios: | | | |
| Output/input | 0.08 | percent | maximum |

AMSC - N/A

FSC 5990

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TABLE I. Requirements - Continued.

| Requirements | Value | Unit | Tolerance |
|---|---|---|---|
| Phase shift : | | | |
| Output/input | 3.5 | degrees | ±1.0 |
| Functional error | ±0.1 | percent | maximum |
| Null voltage @ test voltage: | | | |
| Fundamental | 10 | millivolts | maximum |
| Total | 10 | millivolts | maximum |
| Shift of resolver zero: | | | |
| With voltage | ±1.0 | minutes | maximum |
| With frequency | ±1.0 | minutes | maximum |
| Temperature rise | 15 | degrees C | maximum |
| Mechanical breakaway torque: | 0.2 | oz-in | maximum |
| @ -55°C | 0.4 | oz-in | maximum |
| Radial play | 0.0005 | inches | maximum |
| End play | 0.0010 | inches | ±0.0005 |

NOTE:

1. A voltage value of 0.5 to 26 volts shall be entered in the "volts" designation space on the identification plate.

TABLE II. Part identifying number variant characteristics

| Part Identifying No. 1/ | A ±.010 | B ±.010 | G Maximum | Shaft Style | Terminal End | U ±.0000 -.0005 |
|---|---|---|---|---|---|---|
| M23417/9-01A | .555 | .500 | 2.100 | Spline | Terminal Board | .1872 |
| M23417/9-02A | .305 | .250 | 1.850 | Plain | Wire Leads | .1200 |
| M23417/9-03A | .492 | .437 | 1.850 | Plain | Wire Leads | .1200 |
| M23417/9-04A | .555 | .500 | 1.850 | Plain | Wire Leads | .1200 |
| M23417/9-05A | .305 | .250 | 2.100 | Plain | Terminal Board | .1200 |
| M23417/9-06A | .492 | .437 | 2.100 | Plain | Terminal Board | .1200 |
| M23417/9-07A | .555 | .500 | 2.100 | Plain | Terminal Board | .1200 |

1/ Part identifying number M23417/9-01A shall be the NATO standard.

1/ The suffix letter following the numerical dash number corresponds to the latest modification letter in the type designation.



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TABLE III. Part identifying number cross-reference. 1/

| SUPERSEDED PART NUMBERS | NEW PART IDENTIFYING NUMBERS |
|---|---|
| MIL-R-23417/9 | |
| M23417/9-01 | M23417/9-01A |
| M23417/9-02 | M23417/9-02A |
| M23417/9-03 | M23417/9-03A |
| M23417/9-04 | M23417/9-04A |
| M23417/9-05 | M23417/9-05A |
| M23417/9-06 | M23417/9-06A |
| M23417/9-07 | M23417/9-07A |

1/ All line items shown in Table III refer to equivalent and interchangeable electrical resolvers of the same type designation modification. Part number changes do not affect form, fit or function of the electrical resolvers listed therein.

Custodians:
Army - AR
Navy - AS
Air Force - 85

Review activities:
Navy - OS, EC
Air Force - 17, 99
DLA - ES

User activities:
Army - MI, ME, AV, AT
Navy - OG, MC
Air Force - 14, 19

Preparing activity:
Army - AR

(Project 5990-0380)



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